

IP Services

CCNA 200-301 V1.1

MODULE 6

Topics

6.1 Remote Control Services

6.2 File Transfer Services

6.3 Monitoring Services

6.4 Infrastructure Services

6.5 Quality of Service

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6.1 Remote Control Services

CCNA 200-301 v1.1

Module 6

Telnet

SSH

HTTP

Telnet

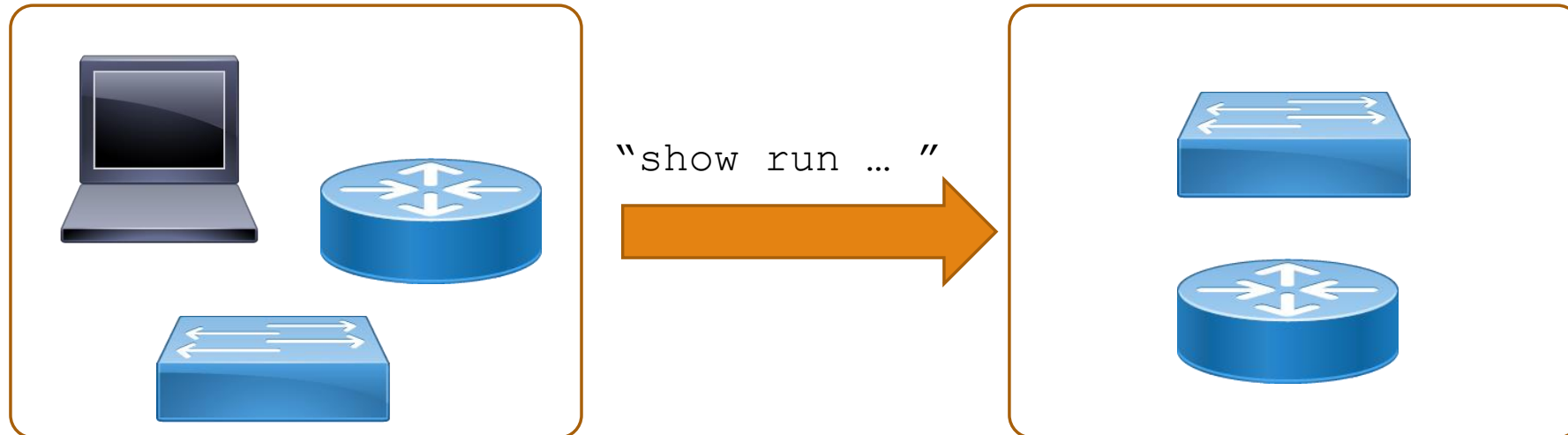
Obtain a command prompt on a remote device

Can be performed in USER or PRIVILEGED EXEC

TCP 23

Unencrypted

Cisco devices can be a telnet client or server



Configure a Telnet Server Example

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# hostname R1
```

```
R1(config)# interface GigabitEthernet0/0
```

```
R1(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
R1(config-if)# no shutdown
```

```
R1(config)# username admin privilege 15 secret LetMeIn
```



Configure a Telnet Server Example (cont'd)

```
R1(config)# line vty 0 15
```

```
R1(config-line)# login local
```

```
R1(config-line)# transport input telnet
```

```
R1(config-line)# end
```

```
R1# copy running-config startup-config
```

```
R2> telnet 192.168.1.1
```



Telnet Commands

Command	Description
<code>line vty 0 4</code> <code>line vty 0 15</code>	Enter vty configuration mode for the desired number of vty telnet sessions (5, or 16)
<code>login local</code>	Username and password is checked against the local database
<code>transport input telnet</code>	Allow telnet connections
<code>telnet <target device IP></code>	Start a telnet session from another Cisco device
<code>CTRL+6+SHIFT, x</code>	Toggle away from the telnet session back to your local CLI
<code>ENTER</code>	Resume your telnet session
<code>exit</code>	End a telnet session
<code>show session</code>	Shows your active outbound telnet sessions to other devices
<code>show users</code>	- Shows active inbound telnet sessions to this device - Shows username, telnet and SSH incoming connections



SSH



Secure Shell (SSH)

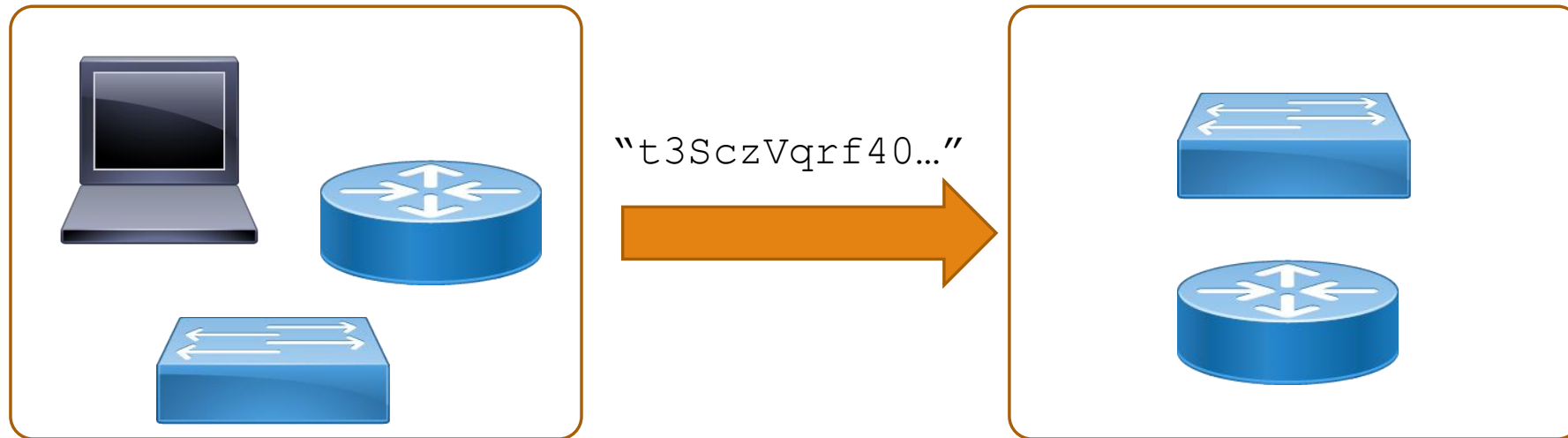
Encrypted replacement for Telnet

Preferred management connection type for Cisco devices

Can be performed in USER or PRIVILEGED EXEC

TCP port 22

Includes SCP (Secure Copy) and SFTP (Secure FTP)



Configuring an SSH Server Example

```
R1> enable
```

```
R1# configure terminal
```

```
R1(config)# hostname R1
```

```
R1(config)# ip domain-name mydomain.com
```

```
R1(config)# crypto key generate rsa
```

```
How many bits in the modulus [512]: 2048
```

```
R1(config)# ip ssh version 2
```

```
R1(config)# username admin privilege 15 secret letmein
```



Configuring an SSH Server Example (cont'd)

```
R1(config)# line vty 0 15
```

```
R1(config-line)# transport input ssh
```

```
R1(config-line)# login local
```

```
R1(config-line)# exit
```

```
R1(config)# ip ssh time-out 60
```

```
R1(config)# ip ssh authentication-retries 3
```

```
R1(config)# end
```

```
R1# copy running-config startup-config
```

```
R2# ssh -l admin 192.168.1.10
```



SSH Commands

Command	Description
hostname	Configure a hostname for the device
ip domain-name <domain name>	Provide the device domain name
crypto key generate rsa	When prompted, choose between 360 – 4096 bit key length
ip ssh version 2	Set the SSH version
username <user> privilege 15 secret <password>	Create a local user with a secret password who will have admin privilege in the session



SSH Commands (cont'd)

Command	Description
<code>line vty 0 15</code>	Enter vty configuration mode for 16 vty lines
<code>transport input ssh</code> <code>transport input ssh telnet</code>	Enable SSH only Enable SSH and if that fails, permit telnet
<code>login local</code>	Require local authentication for the user
<code>ip ssh time-out 60</code>	The user must successfully authenticate in 60 seconds
<code>ip ssh authentication-retries 3</code>	The user can retry authentication 3 times before the session is terminated and they must start over



SSH Connection Commands

Command	Description
<code>ssh -l <username> <destination IP></code>	Start an SSH connection from one Cisco device to another
<code>CTRL+SHIFT+6, x</code>	Toggle back to your local CLI
<code>ENTER</code>	Resume your SSH session
<code>exit</code> <code>logout</code>	End your SSH session
<code>show ssh</code>	View active SSH sessions in our out of this device
<code>show users</code>	- Shows active inbound SSH sessions to this device - Shows username, SSH and telnet incoming connections



HTTP



HTTP Server Services

Server:

Some Cisco devices offer HTTP services for browser or REST API-based remote management

- ISR/ASR routers, Nexus switches, ASA/NGFW Firepower firewalls, WLCs, CUCM, Small business devices
- You can use a browser from your computer to connect

Client:

Most modern Cisco devices can be HTTP clients

- Only for file copying, not remote control



Configure an HTTP Server Example

```
R1# configure terminal
```

```
R1(config)# ip http server
```

```
R1(config)# ip http secure-server
```

```
R1(config)# ip http authentication local
```

```
R1(config)# username [username] privilege 15 secret [password]
```

```
R1(config)# ip http port [port-number]
```

```
R1(config)# ip http secure-port [secure-port-number]
```

```
R1(config)# end
```

```
R1# copy running-config startup-config
```



HTTP Commands

Command	Description
<code>ip http server</code>	Enable the HTTP server
<code>ip http secure-server</code>	Enable the HTTPS server
<code>ip http authentication local</code>	Set authentication to be local
<code>username [username] privilege 15 secret [password]</code> <code>Username admin privilege 15 secret letmein</code>	Create a user that can log in
<code>ip http port [port-number]</code> <code>Ip http port 81</code>	Set the HTTP port
<code>ip http secure-port [secure-port-number]</code> <code>Ip http secure-port 444</code>	Set the HTTPS port

6.2 File Transfer Services

CCNA 200-301 v1.1

Module 6

Copy Command

TFTP

FTP

HTTP

Copy Command

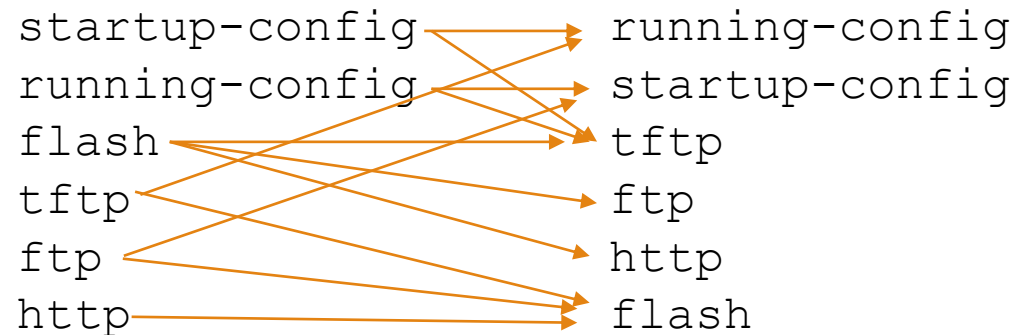
Copy <source> <destination>

You'll be prompted for any missing information before the command is actually executed

- Source file name, destination file name, source IP address, destination IP address, username, password, etc.

You can rename the file as it is copied to the destination

Mix and match any source to any destination (so long as they are configured)



TFTP



Trivial File Transfer Protocol (TFTP)

Transfer files between a Cisco device and a central server

- Configuration, operating system files, updates, backups, logs, other files

UDP port 69

Simpler than other file transfer protocols

- No authentication
- Non-interactive

No browsing the TFTP server directory

- You need to already know the name of the file you want to download

All transmissions are in clear text



Trivial File Transfer Protocol (TFTP) (cont'd)

Cisco devices are typically TFTP clients

Upload/download using the copy command:

- copy tftp flash
- copy flash tftp

If necessary, you can configure a Cisco device to be a TFTP server

- Make sure you have enough storage!

Note: Because TFTP uses UDP with no flow control or error checking, it is not well suited to cross multiple routers or traverse many network segments



Cisco TFTP Examples

Cisco device as TFTP client

- Download a startup-config file from a TFTP server
- `R1# copy tftp startup-config`
- Address or name of remote host `[]? 192.168.1.50`
- Destination filename `[startup-config]? startup-config`

Cisco device as TFTP server

- Making a file available to any TFTP client that requests it – great for copying IOS files between routers
- `R1(config)# tftp-server [file path and name]`
- `R1(config)# tftp-server flash:c2900-universalk9-mz.SPA.151-1.M4.bin`

Verify TFTP server configuration

- `R1# dir flash:`
- `R1# show running-config | include tftp-server`



TFTP Commands

Command	Description
<code>tftp-server [file path and name]</code> <code>tftp-server flash:backup.bak</code> <code>tftp-server flash:c3560- ipservicesk9-mz.150-1.SE.bin</code>	Serve a file as a TFTP server
<code>copy tftp flash</code> <code>copy flash tftp</code>	Download/upload a file from a TFTP server You'll be prompted for the details before the command is executed
<code>copy tftp://192.168.3.47/c3560- ipservicesk9-mz.150-1.SE.bin flash:</code>	You can specify some or all details in one statement You'll be prompted for anything you missed



TFTP Verification Commands

Command	Description
<code>dir flash:</code> <code>show flash:</code>	List all files in your own flash storage including their size
<code>show running-config include tftp-server</code>	Show the currently running configuration Limit output to lines related to TFTP server



FTP



File Transfer Protocol (FTP)

Transfer files between a Cisco device and a central server:

- Configuration, operating system files, updates, backups, logs, other files

More reliable than TFTP

- TCP 21, 20
- No problem going across routers

Session is interactive

Still clear text

Cisco devices are typically FTP clients

Some Cisco devices can be configured as an FTP server

- Be mindful of how much storage the device has!



Configure an FTP Server Example

```
R1> enable
R1# configure terminal
R1(config)# ip ftp server enable
R1(config)# user chrys secret letmein
R1(config)# ip ftp path flash:
R1(config)# end
R1# copy running-config startup-config
R1# dir flash:
R1# show running-config | include ftp
```



Cisco FTP Client Upload Example

```
R1# copy flash ftp
```

```
Source filename [ ]? flash:c2900-universalk9-mz.SPA.151-1.M4.bin
```

```
Address or name of remote host []? 192.168.1.10
```

```
Destination filename [flash:c2900-universalk9-mz.SPA.151-1.M4.bin]?
```

```
User name [ ]? admin
```

```
Password: *****
```



Cisco FTP Client Download Example

```
R1# copy ftp flash
```

```
Address or name of remote host [ ]? 192.168.1.10
```

```
Source filename [ ]? flash:c2900-universalk9-mz.SPA.151-1.M4.bin
```

```
Destination filename [flash:c2900-universalk9-mz.SPA.151-1.M4.bin]?
```

```
User name [ ]? admin
```

```
Password: *****
```



Pre-staging FTP Client Information Example

If desired, you can pre-stage FTP client information to simplify the actual file transfer command

Useful if you perform FTP client operations frequently to the same FTP server

```
R1(config)# ip ftp username chrys
```

```
R1(config)# ip ftp password letmein
```

```
R1(config)# ip ftp server 192.168.1.10
```

```
R1# copy flash:startup-config ftp
```

```
R1# copy ftp:startup-config flash
```



FTP Commands

Command	Description
<code>ip ftp server enable</code>	Enable a Cisco device to be an FTP server
<code>User <name> secret <password></code>	Create a non-privileged user with an encrypted password on the server
<code>ip ftp path <directory-path></code> <code>ip ftp path flash:</code>	Specify the FTP server directory (typically flash:)
<code>dir flash:</code>	List all files on your local flash storage
<code>show running-config include ftp</code>	View current running configuration, limiting output to lines related to FTP
<code>copy flash ftp</code>	Initiate an FTP upload (you will be prompted for info)
<code>copy ftp flash</code>	Initiate an FTP download (you will be prompted for info)



FTP Commands (cont'd)

Command	Description
<code>ip ftp username <name></code> <code>ip ftp username chrys</code>	Pre-stage your FTP client username
<code>ip ftp password <password></code> <code>ip ftp password letmein</code>	Pre-stage your FTP client password
<code>ip ftp server <IP name></code> <code>ip ftp server 192.168.1.10</code>	Pre-stage the FTP server you will connect to
<code>copy flash:<file> ftp</code> <code>Copy flash:startup-config ftp</code>	Upload a file from flash: to FTP
<code>copy ftp flash:<file></code>	Download a file from FTP to flash:



HTTP



HTTP Client Services

Client:

Most modern Cisco devices can be HTTP clients

- Only for file copying, not remote control
- No browser; use **copy** command from terminal

```
R1# copy http [flash|startup-config|running-config]
```

```
R1# copy [flash|startup-config|running-config] http
```

```
R1# copy http://example.com/file.bin flash
```

```
R1# copy flash:file.bin http://example.com/
```



HTTP Client Commands

Command	Description
<code>copy http://<server>/<file> flash</code> <code>copy http://example.com/file.bin flash</code>	Download a file from an HTTP server to local flash storage
<code>copy flash:<file> http://<server>/<file> flash:</code> <code>copy flash:file.bin http://example.com/</code>	Upload a file from an HTTP server to local flash storage

6.3 Monitoring Services

CCNA 200-301 v1.1

Module 6

Syslog

SNMP

Syslog

A standard protocol used by devices to send system log or event messages to a central syslog server

Messages are sent in near real-time

UDP 514

Clear text

Is dependent on each device:

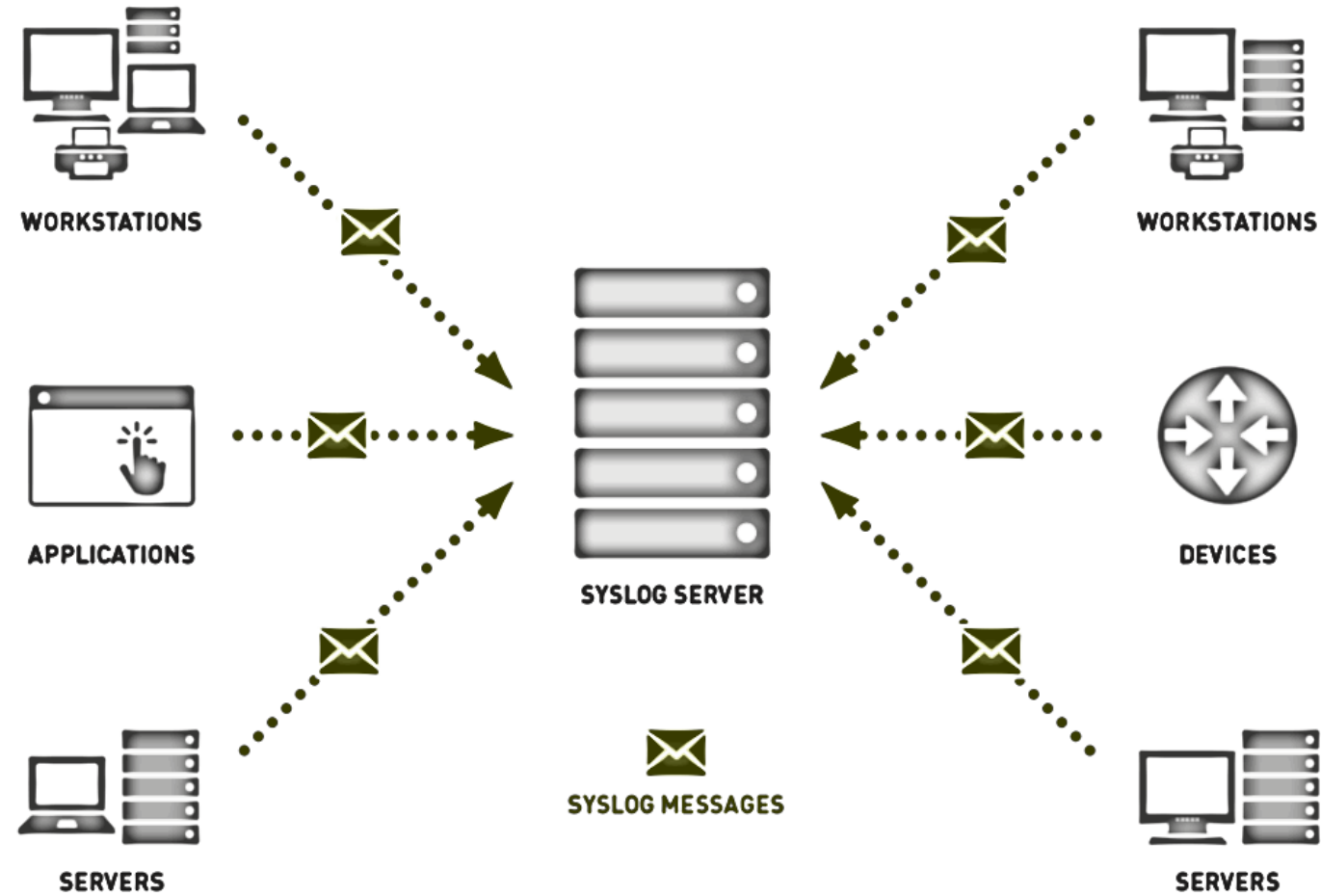
- Logging its own events
- Sending events to the syslog server

Used to collect device log information for:

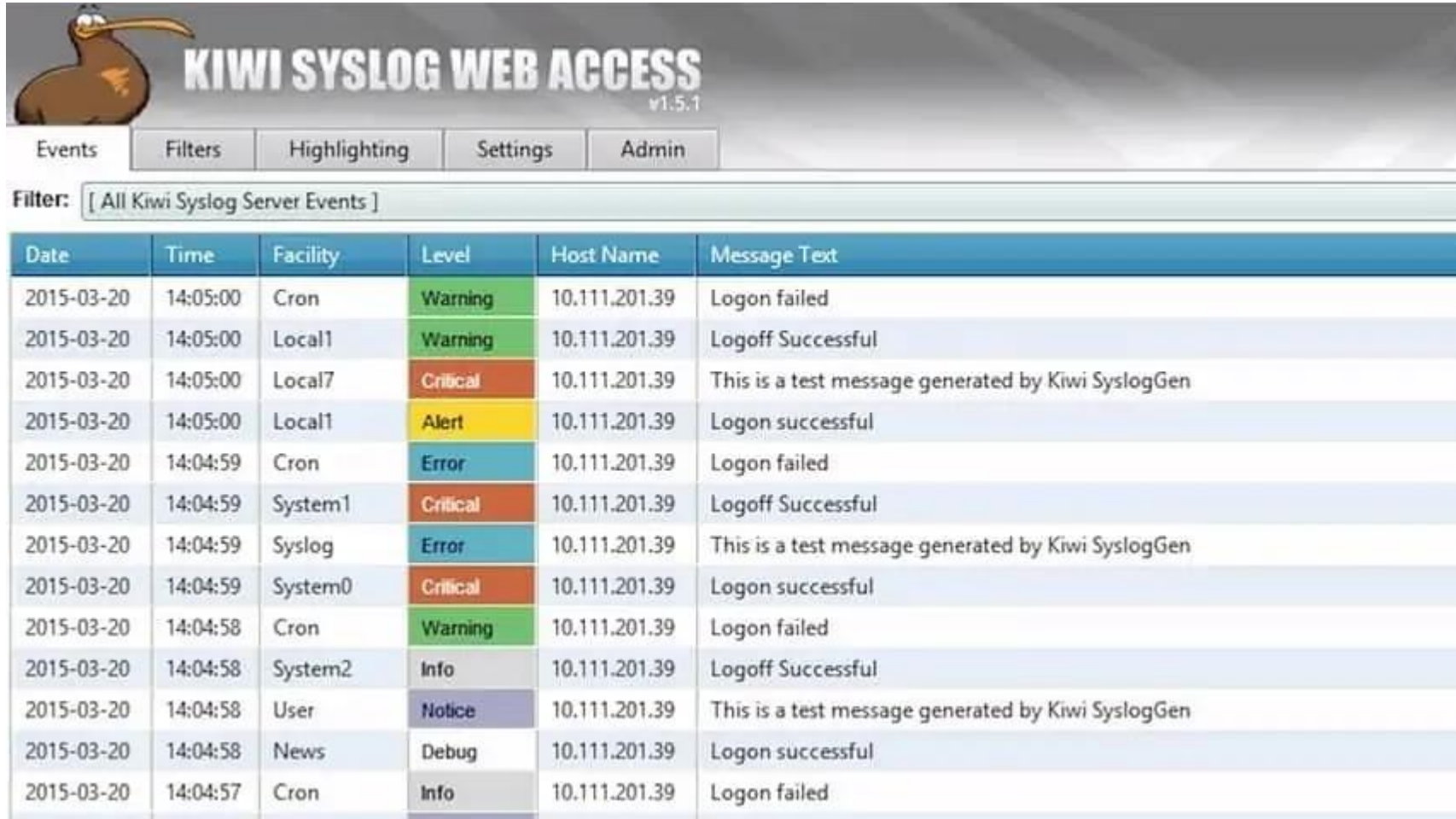
- Device and application monitoring
- Management
- Security auditing



Syslog Example



Syslog Console Example



The image shows the Kiwi Syslog Web Access v1.5.1 interface. At the top left is a cartoon kiwi bird logo. The title "KIWI SYSLOG WEB ACCESS" is in large white letters, with "v1.5.1" below it. Below the title are five tabs: "Events", "Filters", "Highlighting", "Settings", and "Admin". Below the tabs is a filter bar that says "Filter: [All Kiwi Syslog Server Events]". Below the filter bar is a table with 6 columns: "Date", "Time", "Facility", "Level", "Host Name", and "Message Text". The table contains 15 rows of log entries. The "Level" column has color-coded cells: green for "Warning", red for "Critical", yellow for "Alert", blue for "Error", and grey for "Info" and "Notice".

Date	Time	Facility	Level	Host Name	Message Text
2015-03-20	14:05:00	Cron	Warning	10.111.201.39	Logon failed
2015-03-20	14:05:00	Local1	Warning	10.111.201.39	Logoff Successful
2015-03-20	14:05:00	Local7	Critical	10.111.201.39	This is a test message generated by Kiwi SyslogGen
2015-03-20	14:05:00	Local1	Alert	10.111.201.39	Logon successful
2015-03-20	14:04:59	Cron	Error	10.111.201.39	Logon failed
2015-03-20	14:04:59	System1	Critical	10.111.201.39	Logoff Successful
2015-03-20	14:04:59	Syslog	Error	10.111.201.39	This is a test message generated by Kiwi SyslogGen
2015-03-20	14:04:59	System0	Critical	10.111.201.39	Logon successful
2015-03-20	14:04:58	Cron	Warning	10.111.201.39	Logon failed
2015-03-20	14:04:58	System2	Info	10.111.201.39	Logoff Successful
2015-03-20	14:04:58	User	Notice	10.111.201.39	This is a test message generated by Kiwi SyslogGen
2015-03-20	14:04:58	News	Debug	10.111.201.39	Logon successful
2015-03-20	14:04:57	Cron	Info	10.111.201.39	Logon failed



Syslog Facility and Severity

When a syslog message is generated, it is assigned both a facility and a severity level

The combination of these two attributes allows syslog servers and administrators to categorize, prioritize, and respond to log messages

Facility Code + Severity Level = Syslog Priority

<138>Sep 2 14:23:45 RouterA.local2: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to down

Syslog priority is a calculation of facility code, severity level, and format encoding



Syslog Facility Codes

Syslog facilities are pre-defined codes that indicate the origin of a log message

- Identify the source or application generating the message

Standard syslog facilities range from:

- 0 – 15 for various system components
- 16 – 23 for local (custom) use



Standard Syslog Facilities

kern (0): Kernel messages

user (1): User-level messages

mail (2): Mail system

daemon (3): System daemons

auth (4): Security/authorization messages

syslog (5): Messages generated internally by the syslog daemon

lpr (6): Line printer subsystem

news (7): Network news subsystem

uucp (8): UUCP subsystem (Unix-to-Unix Copy Protocol)

cron (9): Clock daemon (cron and at)

authpriv (10): Security/authorization messages (private)

ftp (11): FTP daemon

ntp (12): Network Time Protocol daemon

audit (13): Log audit (used for logging audit records)

alert (14): Log alert (used for logging alert messages)

clock (15): Clock daemon (reserved for system use)

local0 (16): Reserved for local use (custom)

local1 (17): Reserved for local use (custom)

local2 (18): Reserved for local use (custom)

local3 (19): Reserved for local use (custom)

local4 (20): Reserved for local use (custom)

local5 (21): Reserved for local use (custom)

local6 (22): Reserved for local use (custom)

local7 (23): Reserved for local use (custom)



Cisco Syslog Severity Levels

Cisco devices include a severity level as part of their syslog message:

0 – Emergency

1 – Alert, immediate action needed

2 – Critical, could lead to instability or failure

3 – Error, could indicate problems, but do not need immediate attention

4 – Warning, non-critical but should be monitored and addressed

5 – Notification, normal but significant condition

6 – Informational, general operational message, not indicative of any issue

7 – Debugging, detailed messages for troubleshooting and diagnostics, verbose output



Syslog Server Message Responses

You can configure responses on a Syslog server based on facility code, severity level, or both:

Store messages in separate locations

Filter incoming messages to focus only on certain types

Re-route incoming messages to another server or endpoint

Create alerts and actions to respond to certain message types



How Cisco Clients Use Syslog

Cisco syslog client messages by default use facility code local7

If desired, you can also use the other local facility codes 0 – 6 for specific use cases

Messages include the severity level along with facility code



Syslog Client Example

Configure the device to send syslog messages to 192.168.1.100

- Use the default facility local7(23)
- Send all messages that have a severity of informational or higher
- Include a date-time stamp to the millisecond

```
R1# configure terminal
```

```
R1(config)# logging host 192.168.1.100
```

```
R1(config)# logging trap informational
```

```
R1(config)# service timestamps log datetime msec
```

```
R1(config)# logging on
```

```
R1(config)# end
```

```
Router# copy running-config startup-config
```



Syslog Client Custom Facility Example

Configure the device to send syslog messages to 192.168.1.100

- Use a different facility local2
- Send all messages that have a severity of critical or higher

```
R1# configure terminal
```

```
R1(config)# logging host 192.168.1.200
```

```
R1(config)# logging facility local2
```

```
R1(config)# logging trap critical
```

```
R1(config)# logging on
```

```
R1(config)# end
```

```
R1# copy running-config startup-config
```



Configuring a Cisco Syslog Server

Most Cisco devices are already configured to be syslog servers

- You just need to ensure logging is enabled
- You also need to configure other devices to point to the Cisco device as their syslog server
- Keep in mind that Cisco devices have limited resources for storing logs
- Prefer to set up a syslog server on a dedicated server or virtual machine

Default syslog server settings:

- Accept severity level 6 or higher
- Buffer size 4096 bytes
- No filtering or forwarding
- Incoming messages are timestamped



Configuring a Cisco Syslog Server Example

```
R1> enable
```

```
R1# configure terminal
```

```
R1(config)# logging on
```

```
R1(config)# end
```

```
R1# copy running-config startup-config
```



Syslog Commands

Command	Description
<code>logging on</code>	Enable logging (on by default)
<code>logging host <syslog server></code> <code>logging host 192.168.1.200</code>	Specify the syslog server to send messages to
<code>logging facility [local0 - local7]</code> <code>logging facility local2</code>	Optionally specify the syslog facility to use (default is local7)
<code>logging trap <minimal severity level></code> <code>logging trap informational</code>	Optionally specify the minimum severity level to send (default is informational)
<code>service timestamps log datetime msec</code>	Optionally specify if the message should include a timestamp to the millisecond
<code>show logging</code>	



SNMP



Simple Network Management Protocol (SNMP)

Used to monitor and manage network-connected devices in an IP network

- Queries devices on a regular interval (typically every 5 minutes)

Similar to Syslog

Vendor-neutral

UDP 161 and 162

Clear text (unless v3 is used)

Does NOT require the device to maintain its own log

DOES require the device to be able to answer queries



SNMP Components

Manager

- Software running on a server or dedicated appliance
- Queries devices (agents) on a regular interval for their status

Agent

- Software running on a device that responds to SNMP manager queries



SNMP Community Strings

A community string is a text string that acts as a password between the manager and its agents

- The manager must identify itself as belonging to the same “community” as the agent
- Agents will not respond to managers that are configured for a different community string

Read-Only (RO) community string is for queries only

- SNMP manager cannot write or configure anything on the agent device
- Typically set to “public”

Read-Write (RW) community string

- Allows the SNMP manager to both read and modify the settings on the device
- Typically set to “private”

Note: You can change your community strings as desired



SNMP Process

Agents usually wait to be polled by the manager

Agents can also be configured to immediately send an alert (trap) to the manager if a specific event occurs

Common versions include v1, v2c, and v3

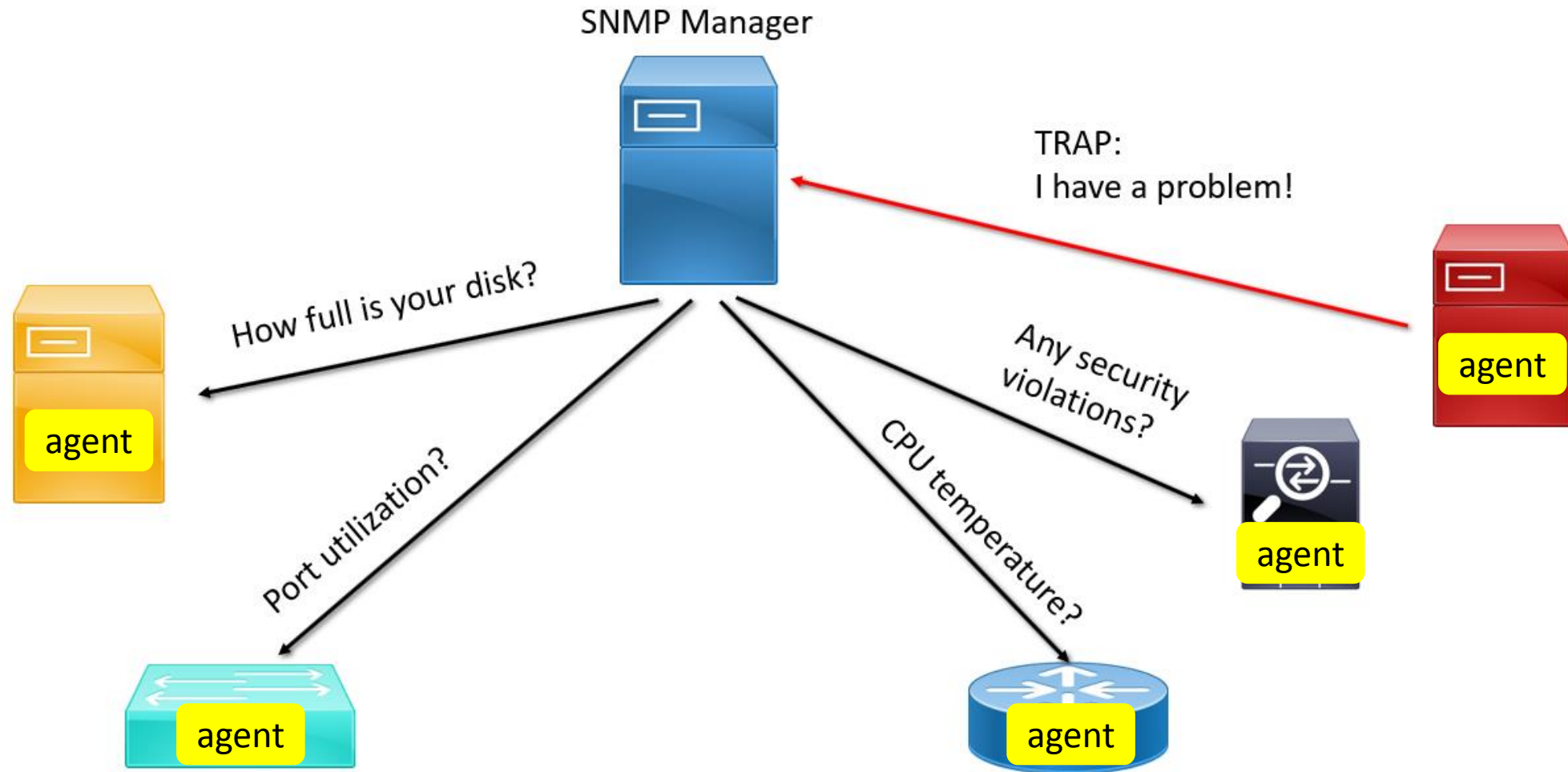
SNMP managers include software such as:

- MRTG, SolarWinds NPM, or OpenNMS, WhatsUp Gold, ManageEngine OpManager

Note: Cisco devices can only be SNMP agents



SNMP Example



Information You Can Enumerate Using SNMP

- Network devices
- Hosts
- Device CPU, RAM and disk utilization
- Users and groups
- Services
- Installed software
- Network shares
- Device configurations
- IP and MAC addresses
- IP and MAC addresses
- ARP tables
- Routing tables
- VLANs
- Port and interface status
- Network traffic
- Security violations
- and much, much more



Configuring SNMP

Enable SNMP and set community strings

- Router(config)# snmp-server community [community-string] [access-level]
- Router(config)# snmp-server community public RO
- Router(config)# snmp-server community private RW

Enable a specific trap

- Router(config)# snmp-server enable traps [notification-type]
- Router(config)# snmp-server enable traps linkup linkdown

Specify manager to send trap to

- Router(config)# snmp-server host <ip address> <community string> [notification-type]
- Router(config)# snmp-server host 192.168.1.100 public linkup linkdown



SNMP Commands

Command	Description
snmp-server community [community-string] [access-level] snmp-server community public RO snmp-server community private RW	Enable SNMP Set the community string “public” for read-only Set the community string “private” for read-write
snmp-server enable traps [notification-type] snmp-server enable traps linkup linkdown	Enable a particular trap
Router(config)# snmp-server host <ip address> <community string> [notification-type] Router(config)# snmp-server host 192.168.1.100 public linkup linkdown	Specify the SNMP manager to send the trap to Include community string and notification type
show snmp	Display SNMP configuration including statistics
show running-config include snmp	Display current running configuration (SNMP only)

6.4 Infrastructure Services

CCNA 200-301 v1.1

Module 6

DHCP

DNS

NTP

Dynamic Host Configuration Protocol (DHCP)

An automated way to assign IP addresses to hosts on a network

Based on the earlier BOOTP protocol

Client issues a Layer 2 broadcast to request an IP address from any listening DHCP server

Server has pre-configured pool of available IP addresses

Server “leases” an address for a limited time to the client

Communications are in clear text with no authentication

Server port = UDP 67

Client port = UDP 68



DHCP Process

AKA “DORA”

Layer 2 Broadcast

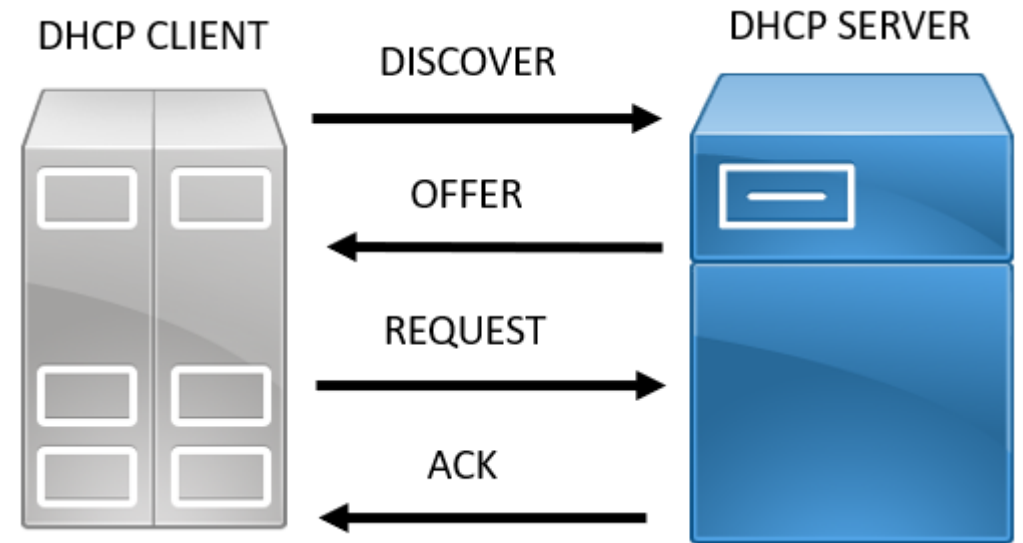
Lease can be limited time or indefinite

Lease will include:

- IP Address
- Subnet Mask
- Length of time for the lease

Lease can include options:

- Default Gateway
- DNS Server(s)
- DNS Domain Name
- Other options



DHCP Relay Agent/IP helper

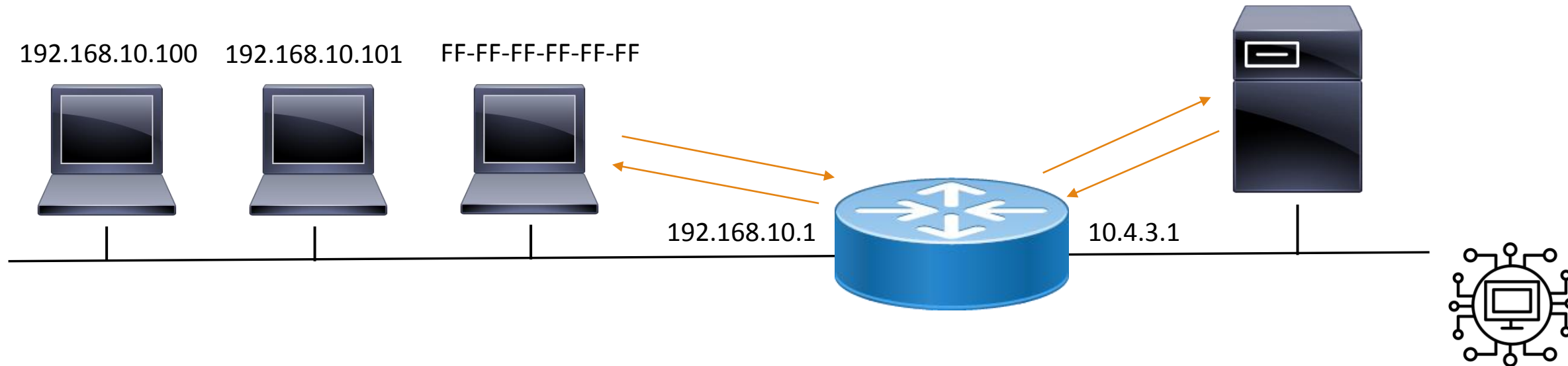
A hardware device or software program that can pass DHCP or BOOTP messages between DHCP clients and servers

- Cisco IP helpers use UDP to carry the DHCP messages

Necessary if the DHCP server is on a different subnet from its clients

- Routers do not pass broadcasts

RFC 1542-compliant routers can be configured as DHCP relay agents



Configure a Cisco DHCP Client Example

Configure on the interface that should be the DHCP client

```
Router(config-if) # ip dhcp
```



Configure a Cisco DHCP Relay Agent Example

Configure on the interface that faces the clients

```
Router(config-if)# ip helper-address < DHCP Server IP >
```

```
Router(config-if)# ip helper-address 192.168.1.1
```



Configure a Cisco DHCP Server Example

1. Exclude IP range from the DHCP scope

```
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

2. Create a pool (DHCP scope)

```
Router(config)# ip dhcp pool MY_DHCP_POOL
```

3. Specify the range of addresses for the pool

```
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
```



Configure a Cisco DHCP Server Example (cont'd)

4. Set the DHCP default gateway scope option

```
Router(dhcp-config)# default-router 192.168.1.1
```

5. Set the DNS servers scope option

```
Router(dhcp-config)# dns-server 8.8.8.8 8.8.4.4
```

6. Set the DHCP lease time (in days)

```
Router(dhcp-config)# lease 7
```

7. Exit and save

```
Router(dhcp-config)# end
```

```
Router# copy running-config startup-config
```



DHCP Commands

Command	Description
<code>ip dhcp</code>	Enable an interface to be a DHCP client
<code>ip helper-address <DHCP Server IP></code> <code>ip helper-address 192.168.1.1</code>	Configure an interface to be a DHCP relay
<code>ip dhcp excluded-address <start> <end></code> <code>ip dhcp excluded-address 192.168.1.1 192.168.1.10</code>	(Global config mode) Exclude any addresses in the subnet from being offered to a DHCP client
<code>ip dhcp pool <name></code> <code>ip dhcp pool MY_DHCP_POOL</code>	Create a DHCP pool (scope)
<code>network <subnet ID> <subnet mask></code> <code>network 192.168.1.0 255.255.255.0</code>	(DHCP config mode) Specify the subnet ID and mask that the leased addresses will come from
<code>default-router <gateway IP></code> <code>default-router 192.168.1.1</code>	Specify the default gateway scope option
<code>dns-server <IP> <IP></code> <code>dns-server 8.8.8.8 8.8.4.4</code>	Specify the DNS scope option
<code>lease <days></code> <code>lease 7</code>	Specify the DHCP lease length



DNS



The Role of DNS in a Network

Maps IP addresses to “friendly” host names

Exists for human convenience

Allows IP addresses to change

Places all organizations and their records in a single hierarchy (DNS namespace)

Uses UDP and TCP port 53

- UDP for queries
- TCP for zone transfers (replication) between servers
- DNS transmissions are in clear text
- DNSSEC uses accompanying digital signatures to guarantee authenticity of a record



Common DNS Record Types

SOA (Start of Authority)

- Indicates the DNS server that is the ultimate authority for the domain

NS (Name Server)

- Indicates this is a DNS server for a specific domain

A

- AKA host record
- IPv4 host record (Name → IP); Example: server1 → 192.168.1.5

AAAA

- IPv6 host record (Name → IP); Example: server1 → 2601:140:8500:d3d0::7723

CNAME

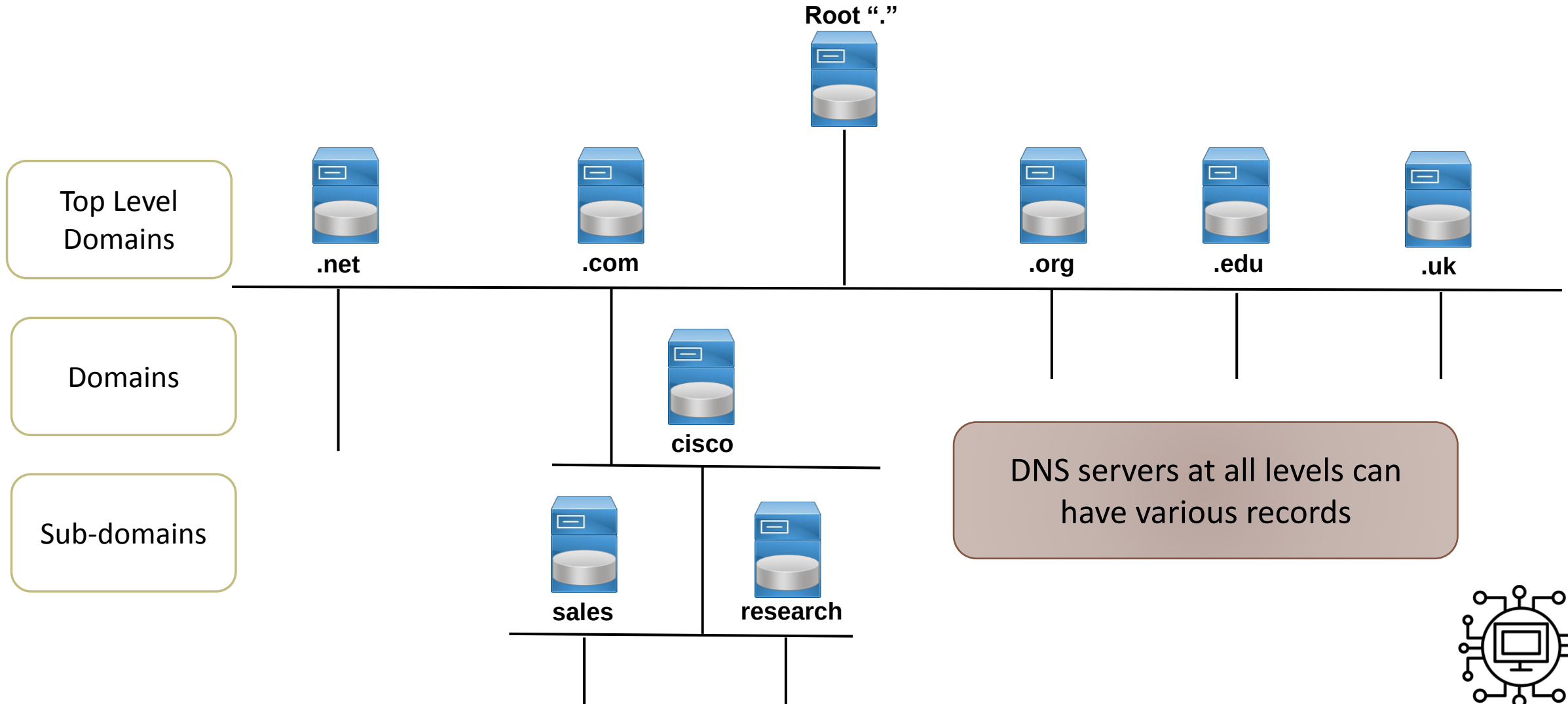
- Alias for a host record; Example: www for server1.company.com

PTR

- Reverse lookup record (IP → Name); Example: 192.168.1.5 → server1



DNS Hierarchy Example



DNS Commands

Command	Description
<code>ip domain-lookup</code>	Enable DNS client / use DNS to resolve names
<code>no ip domain-lookup</code>	Disable DNS client / stop doing name resolution
<code>ip name-server [dns-server-ip]</code> <code>ip name-server 8.8.8.8</code> <code>ip name-server 8.8.4.4</code>	Specify the IP address of your DNS server You can specify one or more DNS servers
<code>ip domain-name [your-domain-name]</code> <code>ip domain-name company.com</code>	Set a default domain name that the router appends to any unqualified hostnames during DNS resolution
<code>ip host [hostname] [ip-address]</code> <code>ip host switch1 192.168.100.100</code>	Create a static host-to-IP address mapping To be used instead of a DNS lookup for that name
<code>ip dns server</code>	Configure the router to be a caching DNS server Can also serve records from <code>ip host</code> entries



DNS Commands (cont'd)

Command	Description
<code>show hosts</code>	Display your DNS cache and host entries
<code>clear host</code>	Clear all dynamically learned DNS addresses
<code>clear host [hostname]</code>	Clear a specific dynamically learned DNS address
<code>ping <hostname></code> <code>ping switch1</code> <code>ping www.cisco.com</code>	See if name resolution is performed before ping
<code>debug ip dns</code>	Perform real-time debugging of DNS processes
<code>undebug all</code>	Stop all debugging



NTP



Network Time Protocol (NTP)

Used to synchronize device clocks on a network

- Servers, clients, Cisco devices, IDS/IPS, mobile devices, etc.

Time synchronization on a network is necessary for many reasons including:

- Consistency across devices, process synchronization, authentication, authorization, auditing, logging, compliance, troubleshooting

UDP 123



NTP Stratum

NTP is organized into a hierarchy of stratum

- Levels 0 – 15
- Stratum 0 is the highest; devices at this level actually generate the reference time

Synchronization is one-way

- Stratum 0 → Stratum 1 → Stratum 2 → Stratum 3, etc.
- The lowest levels, being farthest removed from Stratum 0, will be the least precise
 - Farthest off the original time

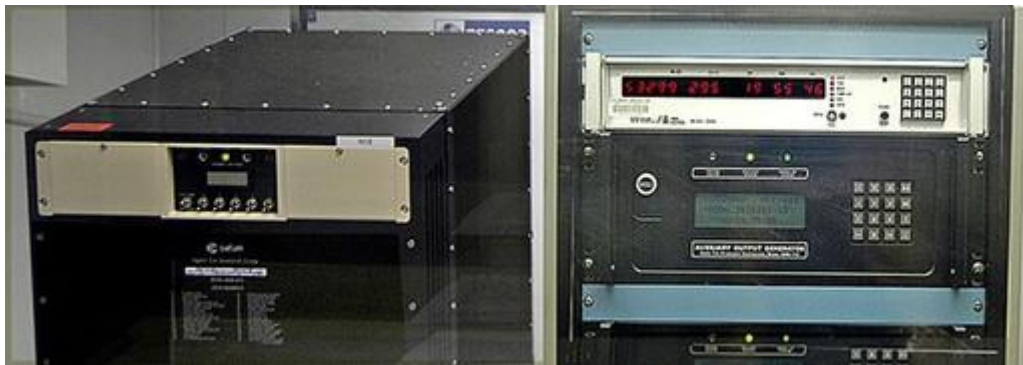


NTP Stratum Infrastructure



NIST Cesium Fountain Atomic Clock

The U.S. Naval Observatory Wash. DC Master Clock



Stratum 0

Actual time sources
Not on the network

Stratum 1

Computers attached to Stratum 0
Master clocks on the network
Distribute time to lower strata

Stratum 2

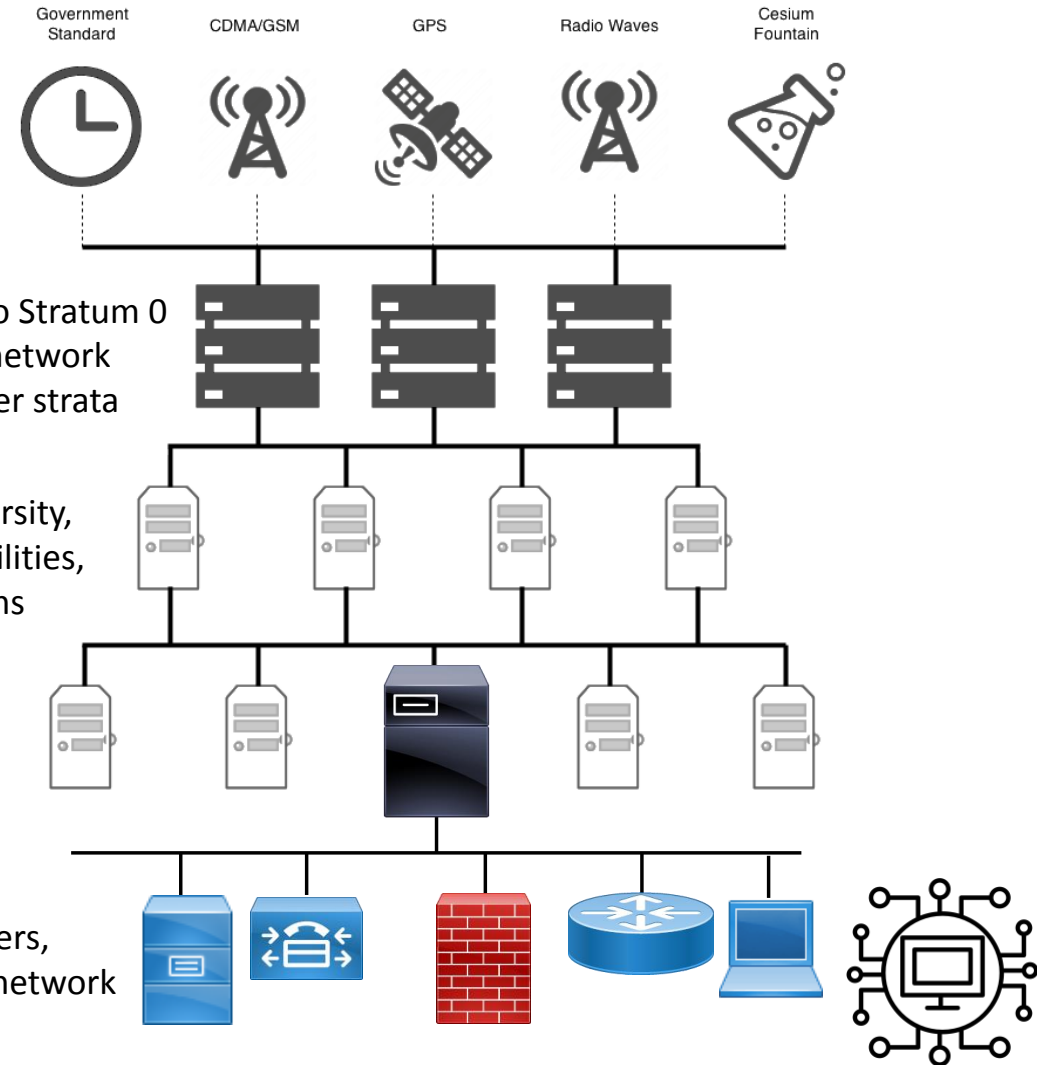
Public servers at university,
research and gov't facilities,
ISPs, large organizations

Stratum 3

Your company's
time server

Stratum 4

Network devices, servers,
workstations on your network



Configuring NTP

You can configure your Cisco device to be an NTP:

- Client
- Server
- Peer (one of two servers that synchronize bi-directionally for fault tolerance)

Peers configuration includes:

- Authentication between the peers
- Shared secret key peers use to authenticate to each other



NTP Client and Server Commands

Command	Description
<code>ntp server [IP address hostname]</code> <code>ntp server 192.168.1.5</code>	Configure the device to be an NTP client Specify the server to synchronize your clock to
<code>ntp master <1-15></code> <code>ntp master 3</code>	Configure the device to be an NTP server Specify the stratum level (usually 2 – 15)



NTP Peer Commands

Command	Description
<code>ntp peer <Peer IP> key <key number></code> <code>ntp peer 192.168.1.2 key 1</code> <code>ntp peer 192.168.1.1 key 1</code>	Declare an NTP peer, with shared secret key number Peer 1 declares peer 2 Peer 2 declares peer 1
<code>ntp authenticate</code>	Require your NTP peer to authenticate to you when synchronizing (must be done on both sides)
<code>ntp authentication-key <key number></code> <code><hashing type> <key value></code> <code>ntp authentication-key 1 md5 123456</code>	Specify the shared secret your peer must use when authenticating to you (must be done on both sides)
<code>ntp trusted-key <key number></code> <code>ntp trusted-key 1</code>	Mark the shared secret as trusted Do on both sides or authentication will fail



NTP Show Commands

Command	Description
<code>show ntp associations</code>	Displays your server or peer IP address, the server or peer's stratum level, your polling interval, server/peer reachability, delays and more
<code>show ntp status</code>	Display whether your device is synchronized by server or peer and its IP address, your own stratum level, your polling interval, your precision level, your local clock as synchronized by NTP, and more
<code>debug ntp packets</code>	Debug all ntp communications
<code>undebug all</code>	Stop all debugging

6.5 Quality of Service

CCNA 200-301 v1.1

Module 6

What is Quality of Service (QoS)?

Differentiated Services Code Point (DSCP)

Per-Hop Behavior (PHB)

What is Quality of Service (QoS)?

Aka traffic shaping

Helps manage packet loss, delay and jitter on your network infrastructure

Ensures that different types of network traffic receive the appropriate priority and resources

Gives priority to traffic that needs it most

- Realtime traffic such as voice and video

To be meaningful, must be supported by every device (switch, router) along the packet's path

- Impossible to enforce on the Internet



Differentiated Services Code Point (DSCP)

A way to identify and mark traffic priority level

- Allows higher priority traffic to receive preferential treatment

Marking placed in Layer 3 packet header

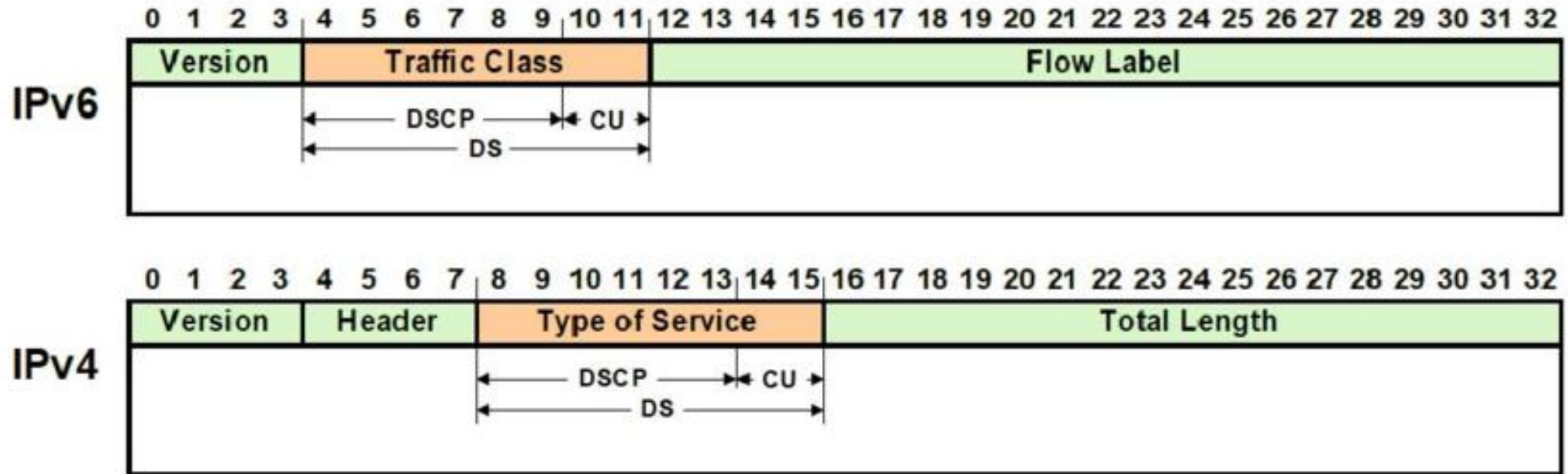
- Various applications can be marked differently
- Range of 0 (lowest) to 63 (highest) priority

Enforced by routers

- Packets with different priorities are placed in different outbound queues



DSCP in the IP Header



DS – Differentiated Service , DSCP – Differentiated Service Code Point, CU – Currently Unused

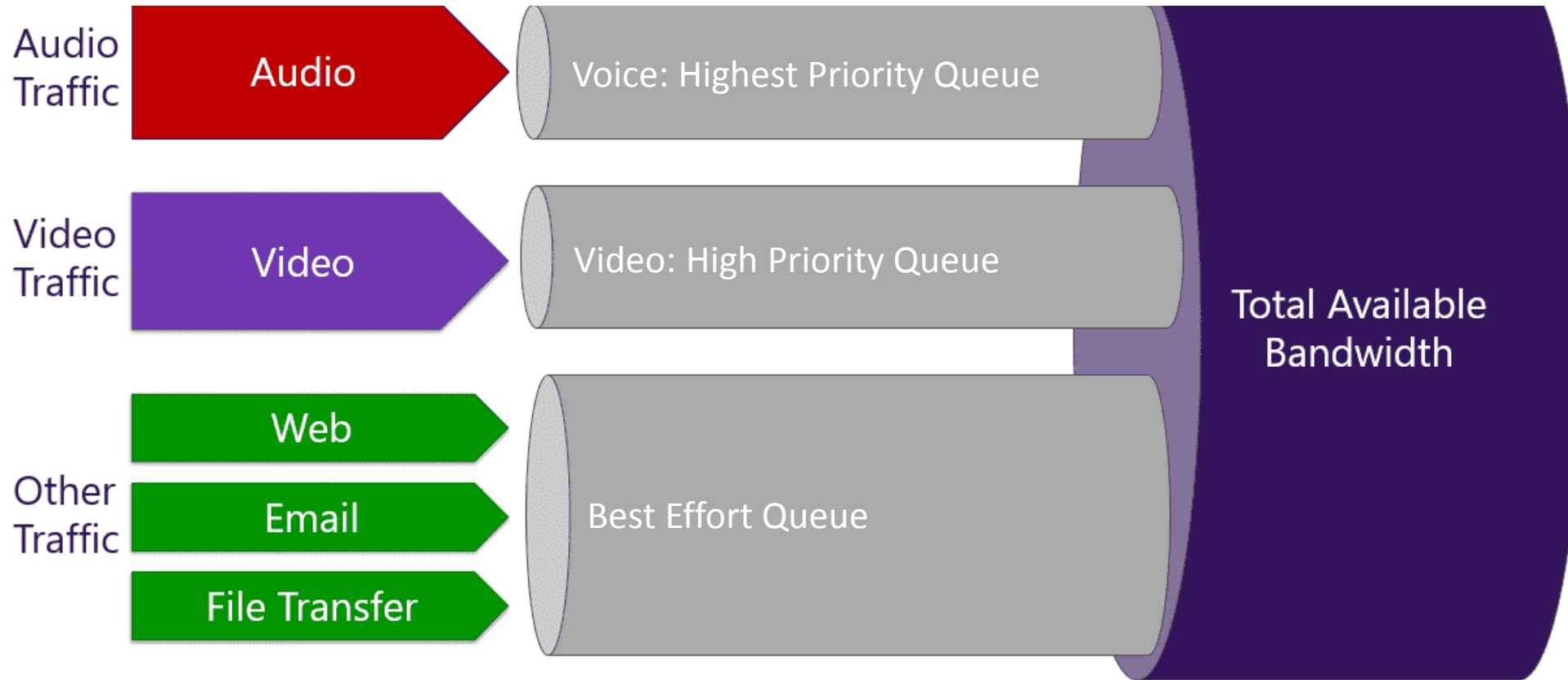


Cisco Baseline DSCP Recommended Values

Application	DSCP Value	Description
Routing	48	Network control
Voice	46	VoIP telephony
Interactive video	34	Multimedia conferencing
Streaming video	32	Multimedia streaming
Mission critical data	26	Defined by organization
Call signaling	24	SIP, H.323
Transactional data	18	Low-latency data
Network management	16	Operations/administration
Bulk data	10	High-throughput data
Scavenger	8	Low priority data
Best effort	0	whatever



DSCP Example Queues on a Router



Per-Hop Behavior (PHB)

A common approach to implementing QoS in IP networks

- QoS defines the policy, PHB implements it

Defines the behavior that each router or switch applies to a packet as it traverses the network

Based on the DSCP value set in the packet's IP header

Note: “Forwarding PHB” is a term you may sometimes hear. It refers to when a router physically forwards a packet based on PHB settings.



PHB Key Components

Component	Description
Classification	Identify and categorize packets based on: Source/destination IP, protocol type, port numbers, or application type
Marking	Tag packets with DSCP levels to indicate their priority
Queuing	Store packets in different outbound (egress) queues based on priority
Congestion Management	How to handle packets during network congestion Drop, delay, or proactively manage
Policing	Drop or re-mark packets that exceed the traffic limit for their priority level
Shaping	Used with marking and other settings for bandwidth control Buffer and delay packets to smooth out bursts of traffic Helps prevent congestion by pacing the traffic flow Unlike policing, holds excess traffic in a queue; releases it at a configured rate

Review

6.6 Review

CCNA 200-301 v1.1

Module 6

Review

Telnet is used to obtain a command prompt on a remote device

It is convenient and easy to use, but not secure

Cisco devices can be a telnet client or server

SSH is an encrypted replacement for Telnet

Some Cisco devices offer HTTP services for browser or REST API-based remote management

Most modern Cisco devices can be HTTP clients, but you must use a command prompt



Review (cont'd)

TFTP is a lightweight protocol for transferring files between a Cisco device and a central server

It is non-interactive, sent in clear text, with no authentication

Using UDP 69, it is best suited for servers that are on the same network segment as the device

All Cisco devices can be TFTP clients

It is possible to configure a Cisco device to be a TFTP server, but you must ensure it has enough storage



Review (cont'd)

FTP is more reliable than TFTP for transferring files

It is interactive, authenticates users, but still sent in clear text

All Cisco devices can be FTP clients

Some Cisco devices can be configured to be FTP servers



Review (cont'd)

Syslog is a standard protocol used by devices to send system log or event messages to a central syslog server

It is dependent on devices being able to log their own events

Messages are sent in clear text on UDP 514

A syslog facility is a pre-defined code that indicates the origin of a log message

Standard syslog facilities range from 0 - 15 for system components; 16 - 23 for local (custom) use

Cisco uses syslog facility local7(23) by default

You can also configure a Cisco device to use other local facilities (local0 - 6) for various use cases



Review (cont'd)

In addition to the facility code, Cisco devices include a severity level as part of their syslog message

The Cisco severity levels range from 0 - 7 (emergency - debugging)

By default, Cisco sends syslog messages for severity level 6 or higher

You can configure different responses on a Syslog server based on message facility code, severity level, or both

Most Cisco devices are already configured to be syslog servers

All Cisco devices can be configured to be syslog clients



Review (cont'd)

SNMP is similar to syslog

Unless you use SNMP v3, messages are sent in clear text on UDP 161 and 162

Unlike syslog, SNMP does not require the device to maintain its own log

An SNMP manager regularly polls devices running the SNMP agent for status updates

The SNMP manager must use the same community string as the agent

Otherwise the agent will not respond

The two common SNMP community strings are public, for read-only operations, and private, for read-write operations



Review (cont'd)

DHCP is an automated way to assign IP addresses to hosts on a network

Client issues a Layer 2 broadcast to request an IP address from any listening DHCP server

The DHCP server has pre-configured pool of available IP addresses that include lease duration, and scope options such as default gateway and DNS server(s)

You can configure a Cisco device interface to be a DHCP client

You can also configure a Cisco device to be a DHCP server

RFC 1542-compliant routers can also be configured as DHCP relay agents



Review (cont'd)

DNS maps IP addresses to “friendly” host names

DNS places all organizations and their records in a single hierarchy called the DNS namespace

At the top of the namespace is the root, followed by top level domains, then domains, and then sub-domains

DNS servers exist at every level of the DNS namespace, and can contain various records there

You can configure a Cisco device to be a DNS client

You can also configure a Cisco device to be a limited DNS server

As a server, the device will query other DNS servers on your behalf, cache the response, and also offer up any requested records from its own static host-to-IP address mappings



Review (cont'd)

NTP is used to synchronize device clocks on a network

NTP infrastructure is organized into a hierarchy of stratum levels, from 0 - 15

Stratum 0 is the highest level; it contains devices that actually generate time such as GPS, Cell towers, Cesium fountains, and other sources

Stratum 1 devices are computers that obtain the time from a Stratum 0 device, and make that time available on the network

Devices at each stratum level synchronize devices at the next lower level

Cisco network devices are typically at Stratum 4

You can configure a Cisco device to be an NTP client, server, or peer

NTP peers exist for fault tolerance, and can synchronize each other bi-directionally



Review (cont'd)

Quality of Service is used to manage packet loss, delay and jitter on your network infrastructure

It gives priority to traffic that needs it most, such as voice and video

To be meaningful, QoS must be supported by every device along the packet's path



Review (cont'd)

Differentiated Services Code Point (DSCP) is a way to identify and mark traffic priority level

DSCP markings are placed in the packet's Layer 3 header, and are enforced by routers that the packet passes through

Packets with different priorities are placed in different outbound queues

DSCP markings range from 0 (lowest) to 63 (highest) priority

Each router that is configured for QoS has various outbound (egress) queues

Each queue has a different priority level

Outbound packets are placed in the appropriate queue based on their DSCP marking



Review (cont'd)

Per-Hop Behavior (PHB) is a common mechanism for enforcing QoS

PHB uses classification, marking, and queuing to categorize, tag, and store packets based on traffic type

PHB then uses congestion management to determine what to do with a packet when there is network congestion

PHB uses policing to drop or re-mark packets that exceed the traffic limit for their priority level

PHB uses shaping to smooth out bursts of traffic

In shaping, packets are not dropped, but are instead stored in a buffer and delayed to be released a little later

